

1. Introduction

VendorFlow is a localized e-commerce platform designed to bridge the digital gap for small businesses. It provides a dedicated digital storefront for local vendors to sell their products directly to their community. The prototype demonstrates a complete, seamless user journey from shop discovery to order placement.

2. Background

Rise of Quick-Commerce: Large platforms (e.g., Blinkit, Instamart) are changing consumer shopping habits. Vendor Struggle: Small, local shops often lack the technical resources or capital to build an independent online presence. The Goal: Level the playing field by offering an accessible, professional, and easy-to-manage digital platform for neighborhood vendors.

3. Core Use Cases

Customer - Shop Discovery: Customers can browse nearby registered shops based on categories. Customer - Ordering: Browsing product listings, managing a cart, placing orders, and viewing order history. Vendor - Onboarding: A seamless registration flow for local business owners to bring their shop online. Vendor - Operations: Managing product inventory, updating order statuses, and viewing sales analytics.

4. Methodology

Frontend-First Approach: Developed an MVP focusing heavily on User Experience (UX) and visual polish. Agile Prototyping: Rapid iteration of UI components to validate the user journey before heavy backend development. Iterative Development: Creating a robust architectural foundation with separate Frontend and Backend ecosystems.

5. Frontend Technologies

Technology	Version	Purpose
React	18	UI component framework
Vite	5.4	Frontend build tool & dev server
Tailwind CSS	3.x	Utility-first CSS styling
React Router	v6	Client-side routing
Axios	1.x	HTTP client for API calls

6. Backend & Database Technologies

Technology	Version	Purpose
Java	17	Core programming language
Spring Boot	3.2.3	Backend application framework
Spring Security	6.x	Authentication & authorization
Spring Data JPA	3.2.3	Database abstraction layer
Hibernate ORM	6.4.4	Object-relational mapping
JWT (Java JWT)	0.12.3	JWT token generation & validation
BCrypt	(Spring Security)	Password hashing
Maven	3.9.6	Build tool & dependency management
Database		
PostgreSQL	17	Relational database
HikariCP	(bundled)	Database connection pooling

7. Result

Customer Marketplace: Delivered a functional shopping experience including shop filtering, a dynamic cart, and checkout flow. Vendor Portal: Created a comprehensive dashboard featuring real-time order management and product tracking. Outcome: A highly interactive, startup-ready prototype validating the core business concept and technical UX requirements.

Admin Dashboard

Manage vendor approvals

[Refresh](#) Pending Approval (0)

No vendors found.

 Approved Vendors (2)

Store Name	Type	Location	Reliability	Status
QuickMart	Grocery	28.6139, 77.2090	★ 3.0 / 5	approved
FreshStore	Grocery	28.5355, 77.3910	★ 4.8 / 5	approved

Welcome back

Sign in to your VendorFlow account

Email

Password

[Sign In](#)Don't have an account? [Register](#)

Demo Credentials (password: password123)

 admin@vendorflow.com — Admin
 quickmart@vendorflow.com — Vendor
 freshstore@vendorflow.com — Vendor
 customer@vendorflow.com — Customer